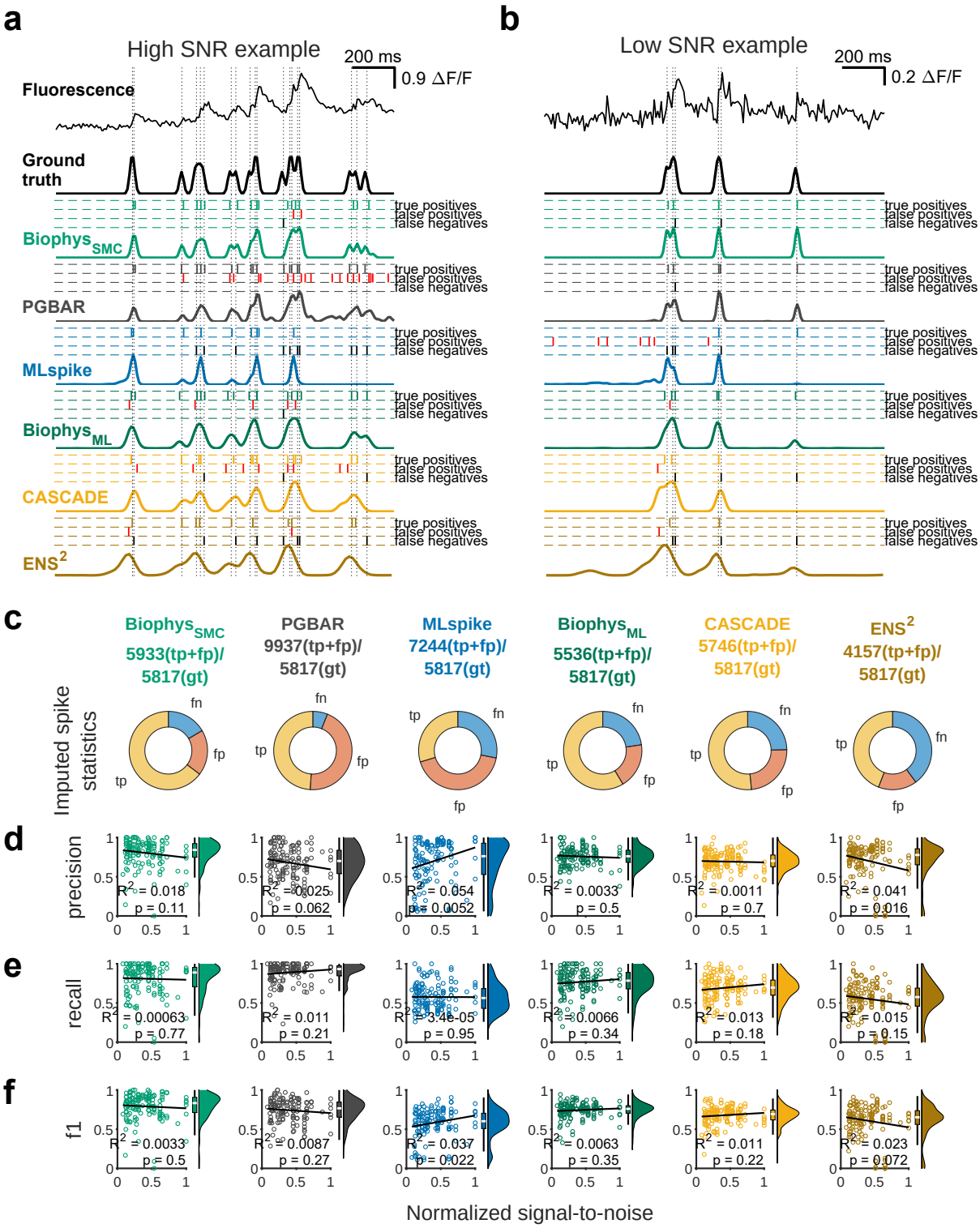

a
High SNR example
200 ms
0.9 ΔF/F
Fluorescence
Ground truth
Biophys SMC
PGBAR
MLspike
Biophys ML
CASCADE
ENS2
true positives
false positives
false negatives
b
Low SNR example
200 ms
0.2 ΔF/F
c
Biophys SMC 5933(tp+fp)/ 5817(gt)
PGBAR 9937(tp+fp)/ 5817(gt)
MLspike 7244(tp+fp)/ 5817(gt)
Biophys ML 5536(tp+fp)/ 5817(gt)
CASCADE 5746(tp+fp)/ 5817(gt)
ENS2 4157(tp+fp)/ 5817(gt)
Imputed spike statistics
tp
fp
fn
d
precision
R2 = 0.018 p = 0.11
R2 = 0.025 p = 0.062
R2 = 0.054 p = 0.0052
R2 = 0.0033 p = 0.5
R2 = 0.0011 p = 0.7
R2 = 0.041 p = 0.016
e
recall
R2 = 0.00063 p = 0.77
R2 = 0.011 p = 0.21
R2 = 3.4e-05 p = 0.95
R2 = 0.0066 p = 0.34
R2 = 0.013 p = 0.18
R2 = 0.015 p = 0.15
f
f1
R2 = 0.0033 p = 0.5
R2 = 0.0087 p = 0.27
R2 = 0.037 p = 0.0022
R2 = 0.0063 p = 0.35
R2 = 0.011 p = 0.22
R2 = 0.023 p = 0.072
Normalized signal-to-noise